

PLEASE ANSWER ALL QUESTIONS IN EACH SECTION

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

(10 MARKS)

- a) Why should you develop both logical and physical Data Flow Diagrams (DFD) for systems? Describe TWO advantages of drawing a logical DFD before a physical DFD for a new information system

(6 marks)

- b) Describe what exploding data flow diagrams (DFD) means.

(2 marks)

- c) “A data flow diagram should **balance** between levels”. Give explanation of the statement based on input and output of a specific system.

(2 marks)

QUESTION 2

(10 MARKS)

World's Trend is a mail order supplier of high-quality, fashionable clothing. Customers place orders by telephone, by mailing an order form included with each catalog, or via the Web site. Summary of Business Activities

- i. When customer orders come in, the item master and the customer master files are both updated. If an item is out of stock, the inventory control department is notified.
- ii. If the order is from a new customer, a new record is created in the customer master file.
- iii. Picking slips are produced for the customer order and sent to the warehouse.
- iv. A shipping statement is prepared.

Question 1

- a) because they provide different views of a system. A logical DFD focus on what the system does such as the data flow, process, input and output. While a physical DFD focus on how the system is implemented like their hardware, software, database and also entities. The 2 advantage are:
- it will help the analyst to understand the actual business process. This will ensure that the system will the user requirement before implementation phase.
 - it will act as a blueprint to make the physical DFD because the required information are identified, the developer can identify the hardware, software and database needed to implement.
- b) breaking down a higher level of DFD into a more detail lower level diagram. For example, breaking down each process in Diagram 0 into child diagram where it shows the subprocess involved.
- c) it means the input and output process at one level must remain the same even expanding into a lower-level DFD

- v. The process of shipping a customer order involves getting the goods from the warehouse and matching up the customer shipping statement, getting the correct customer address, and shipping it all to the customer.
- vi. The customer statement is generated and a billing statement is sent to a customer once a month.
- vii. An accounts receivable report is sent to the accounting department.

A context-level data flow diagram of the above case study is as shown in Figure 1 and a level 0 diagram is shown in Figure 2. The “Add customer order” process, labeled as 1.0 is further exploded into a child diagram shown in Figure 3.0.

- a. Construct the logical DFD child diagram (as shown in Figure 3.0) into a physical Data Flow Diagram.

(10 marks)

- b. Describe the concept of partition. In your opinion, do you need to partition the physical data flow diagram of the “Add customer order” process? Justify your answer.

(5 marks)

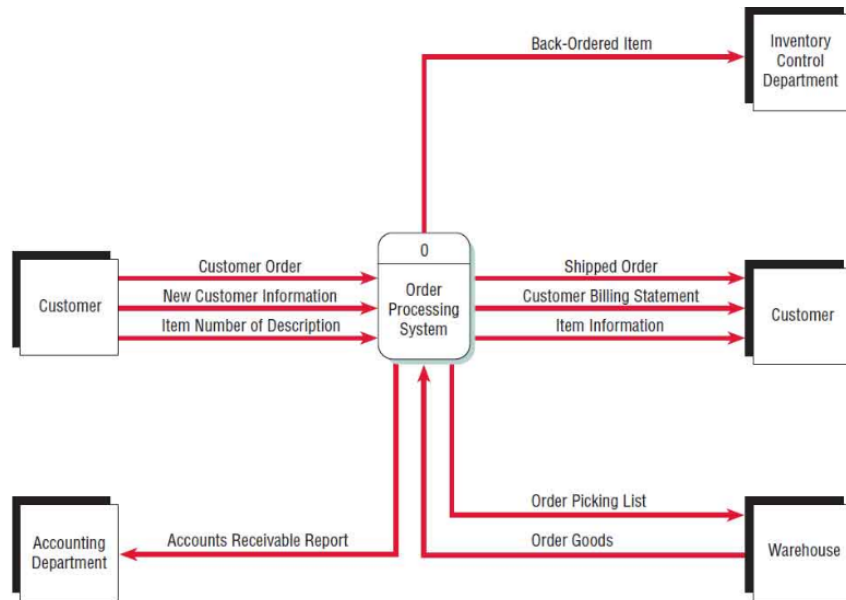


Figure 1: A context-level data flow diagram for the order processing system at
World's Trend

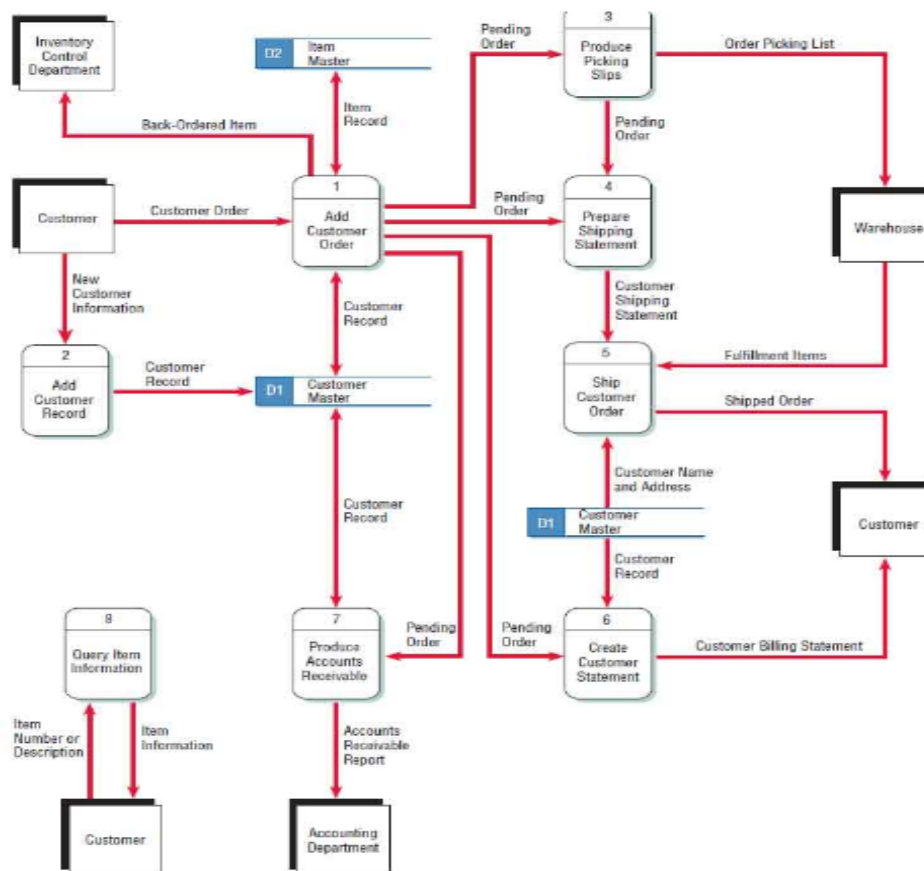


Figure 2: Diagram 0, of the order processing system for World's Trend Catalog Division.

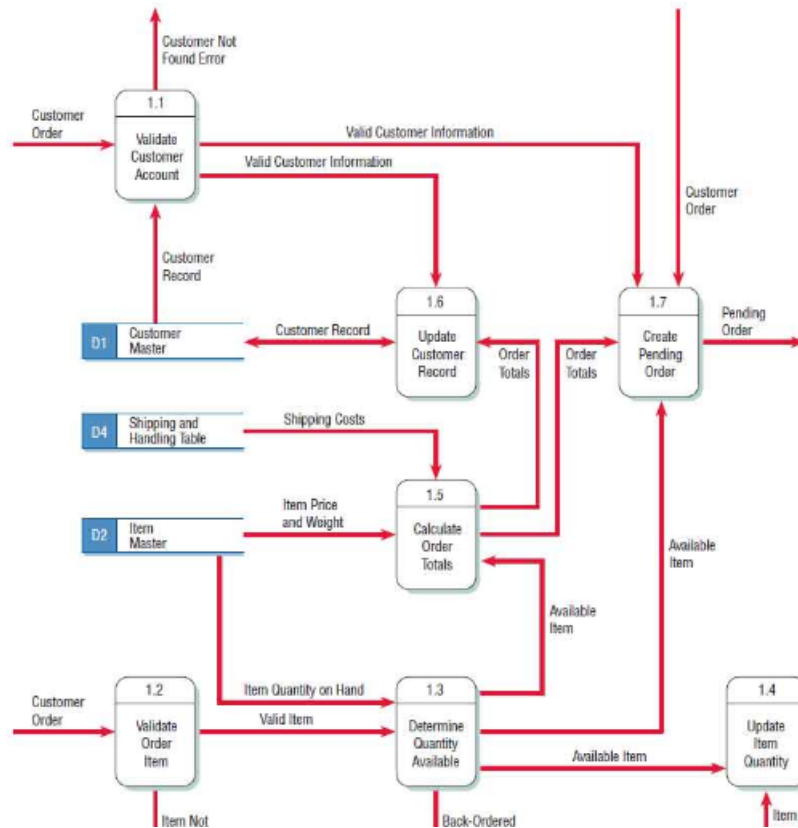


Figure 3: A Logical DFD of the order processing system for World's Trend Catalog Division.

QUESTION 3

(15 MARKS)

Your company has been hired as IT consultant to assist 'Exotic Treat' company in developing a newly system that allows their customer to purchase homemade sweets and cakes through online booking application. One of your job as the in-charge Business Analyst is to assist the development team in analyzing the detail process and logic flows for the determination of total payment invoice. The criteria in calculating the total payment invoice are as follows:

- Customers with e-coupon will be granted for RM10 discounts less for each total payment invoice

- Customers who spent more than RM250 purchase items with:
 - Numbers of order items (more OR less than 4 order items);
 - Choices of delivery day once the order is placed (next day, second day, OR seventh day)

The specific conditions to calculate total payment invoice are summarized using the decision table in **Table 1**.

Table 1: Decision table for calculating shipping charges and total invoice

Total <i>RMpurchase</i> <i>e</i> > RM250	YES						NO					
Number of items (n)	$N \leq 3$			$N \geq 4$			$N \leq 3$			$N \geq 4$		
Delivery day	Nex t	2 nd	7 th	Next	2 nd	7 th	Nex t	2 nd	7 th	Next	2 nd	7 th
DeliveryRMcharge	25	10	n*1. 5	n*6. 0	n*2. 5	Fre e	35	15	1 0	n*7. 5	n*3. 5	n*2. 5
e-Coupon	YES						NO					
Total invoice (RM)	(Total <i>RMpurchase</i> + Delivery <i>RMcharge</i>) – RM10 discounts less						Total <i>RMpurchase</i> + Delivery <i>RMcharge</i>					

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- (a) Construct an optimized decision tree to demonstrate the logic conditions and actions sequences of process flows in calculating the total payment invoice

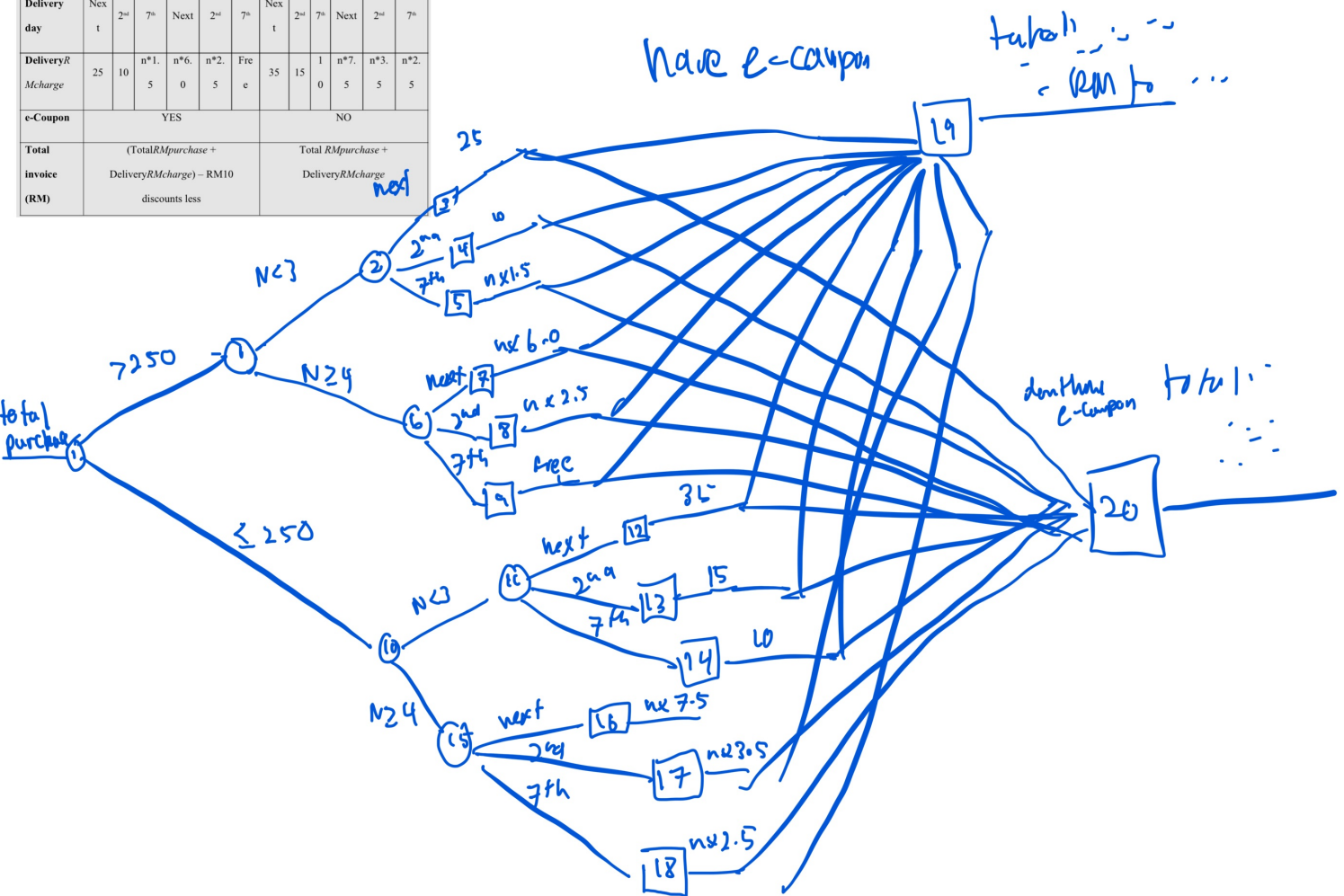
(9 marks)

- (b) In your opinion, which method (decision table or decision tree) is more helpful to analyze the process flows in calculating total payment invoice? Justify your reason.

(6 marks)

b) A decision tree is more helpful for analysing the process because it provides a visual representation of the sequence of decision and action - it allows user to easily follow the condition step-by-step. A decision table can become difficult to understand because it list all possible condition when it can lead to misunderstanding

Total <i>RM</i> purchase $e > RM250$	YES						NO					
Number of items (n)	$N \leq 3$			$N \geq 4$			$N \leq 3$			$N \geq 4$		
Delivery day	Nex t	2 nd	7 th	Nex t	2 nd	7 th	Nex t	2 nd	7 th	Nex t	2 nd	7 th
Delivery/R Mcharge	25	10	n*1. 5	n*6. 0	n*2. 5	Fre e	35	15	1 0	n*7. 5	n*3. 5	n*2. 5
e-Coupon	YES						NO					
Total invoice (RM)	(Total <i>RM</i> purchase + Delivery <i>RM</i> charge) – RM10 discounts less						Total <i>RM</i> purchase + Delivery <i>RM</i> charge <i>note</i>					



QUESTION 4

(10 MARKS)

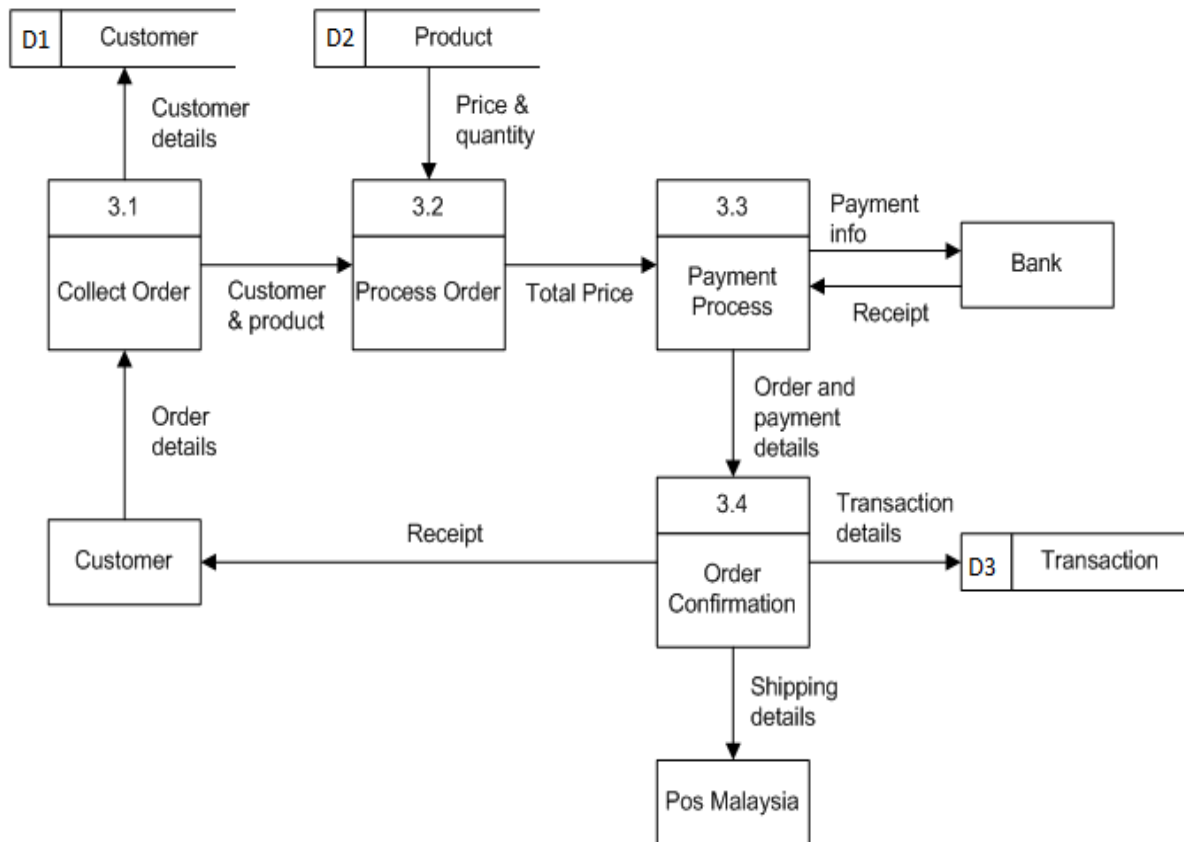


Figure 4: Data Flow Diagram Level 1 of new order process for the Online Shopping System

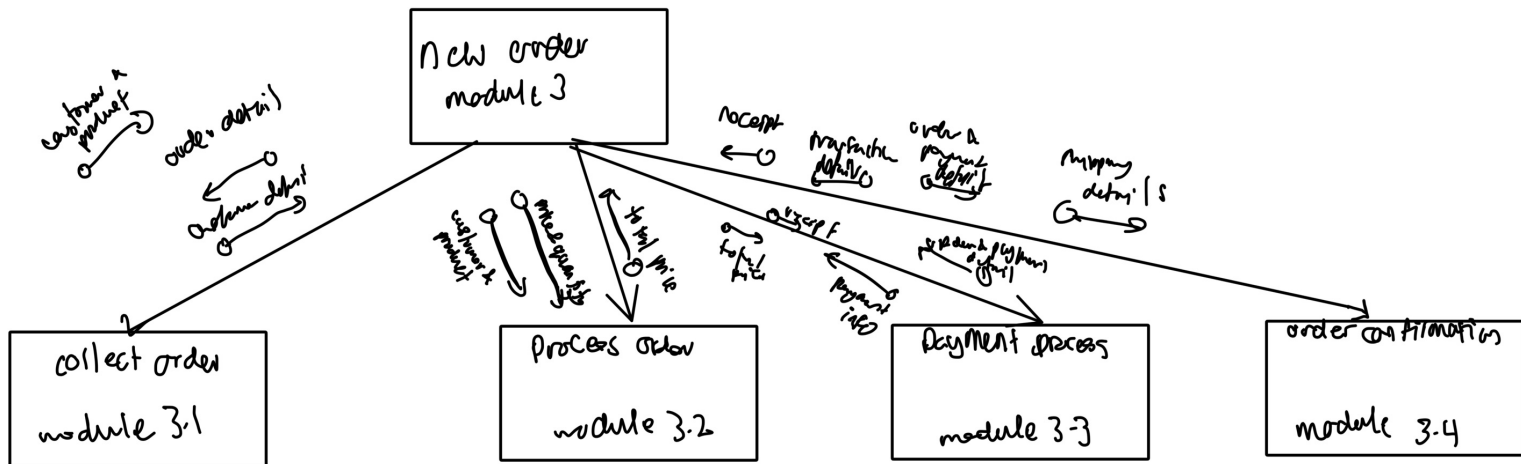
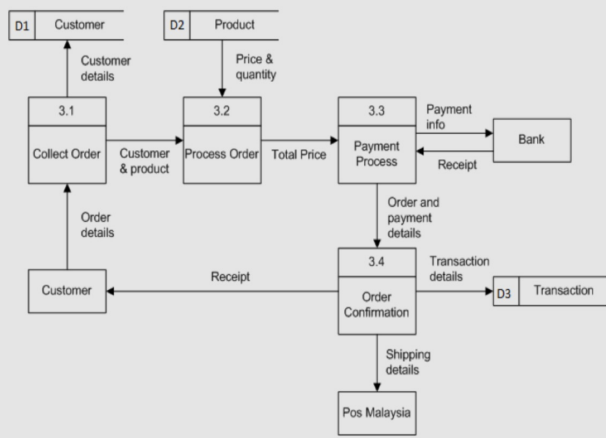
- (a) Figure 4 describes a new order process for the Online Shopping System. Convert the data flow diagram into a structure chart.

(8 marks)



- (b) Analyse the structure chart that you created in (a) and identify a module which has a high degree of cohesion and is loosely coupled. Provide your justification.

(2 marks)



b) collect order is high degree of cohesion and is loosely coup

QUESTION 5

(10 MARKS)

The screenshot shows a Microsoft Access window titled "Microsoft Access - [ISD EXAM : Form]". The form is titled "NEW STUDENT REGISTRATION SYSTEM" and contains the instruction "Please Fill Your Information Completely". The form fields are as follows:

- First Name :
- IC Number: Matric Number:
- Email: Last Name:
- Telephone Number: Example: 6075532421
- Handphone Number:
- Current Address:
- Permenant Address: City:
- Country: Zip Code:
- Sex: Course:
- CPA:

The form is displayed in "Form View" and the status bar shows "Record: 1 of 1". The Windows taskbar at the bottom shows the Start button and several open applications including Yahoo! Messenger, Internet Explorer, Exam Collect..., and Microsoft Word.

Figure 5: The interface of New Student Registration System

- (a) Analyze the interface screen in Figure 5, and identify **FOUR (4)** weaknesses of the input design. Then, redraw a new interface screen for the Figure 5.

(6 marks)

- (b) User interface design focuses on user interaction with the computer, including input design and procedures. Discuss **FOUR (4)** guidelines that can be applied in designing the user interface.

(4 marks)

1. messy layout, boxes aren't lined up properly
2. the fields are in confusing order, where last name are put far from the first name
3. spelling mistake, Permanent Address instead of Permanent Address
4. wrong input box type, fields like sex and course should use dropdown menu

New Student Registration System	
PERSONAL INFORMATION	
First Name :	
Last Name :	
Ic. Number :	
Matric Number :	
Sex :	▼
CONTACT DETAILS	
Email :	
Handphone Number :	
(Format : 601xxxxxxxxx)	
ADDRESS & ACADEMIC INFO	
Current Address :	
Permanent Address :	
City :	Zip Code :
Country :	▼
Course :	▼
CPA :	
Save Submit Close	

- b)-keep the display simple
 - keep the display presentation consistent
 - facilitate the user movement among display screens
 - create an attractive display screen and pages.

QUESTION 6

(15 MARKS)

- (a) User acceptance test is the process whereby actual users test a completed information system through Alpha and Beta testing. Explain briefly **THREE (3)** types of Alpha testing.

(6 marks)

- (b) Given **THREE (3)** short scenarios below, identify the most suitable installation strategy for each scenario and justify your answer.

- i) ABC Commerce Merchant Bank has many branches that cover the whole states of Malaysia. The Bank Management Officers have decided to install a new automated teller machine (ATM) system at each branch office to increase performance over the current ATM systems. The management teams are very concerned about minimizing risks and reducing installation cost.
- ii) Kamal wants to improve the inventory process for his minimarket. He already purchased an inventory control system and needs to install the system quickly. The new system should be running as soon as possible. He does not mind to take the risk and to bear the installation cost for the new system.
- iii) Alpha & Beta Corp. is a semi-conductor multinational company operated in Singapore, Kuala Lumpur, Jakarta and Bangkok. Recently they completed the development of Enterprise Resource Planning (ERP) system to replace the old Manufacturing Resource Planning (MRP) system. The new ERP system is huge

multiple locations

- (a) User acceptance test is the process whereby actual users test a completed information system through Alpha and Beta testing. Explain briefly **THREE (3)** types of Alpha testing.

component testing is the testing of individual hardware or software component

Integration testing is testing to expose defects in the interface and interaction between integrated component of system

system testing is testing an integrated system to verify that it meets specified requirements.

b) i) Parallel installation so that old system and the new system could run at the same time until management decide to turn off the old system

ii) Direct installation so that the new system could ^{be} used directly without care to take the risk and cost

iii) Single location installation so that the new system could be brought out at one location without risking the other system at different location until it is decided to change the new system using the experience at one side.

and covers several departments (manufacturing, finance, & procurement). Since the project is very expensive, the management teams are concerned about minimizing risks of installation failure.

(6 marks)

- (c) Why does the implementation of some systems failed? Give **TWO (2)** opinions about this situation.

(3 marks)

c) Lack of management support because if management does not actively support the new system, employees will not receive a clear instruction of the system

→ Insufficient commitment to change because if user and stakeholders are not genuinely committed to adapt to new system, therefore the new system will be irrelevant.

SECTION B: CASE STUDY

[30 MARKS]

INSTRUCTION: READ THE FOLLOWING SCENARIO AND ANSWER

QUESTIONS (a) UNTIL (c).

Government Solutions Company (GSC) sells computer equipments to federal government agencies. Whenever a federal agency needs to purchase equipment from GSC, it issues a purchase order against a standing contract previously negotiated with the company. GSC holds several standing contracts with various federal agencies. When a purchase order is received by GSC's contracting officer, the contract number referenced on the purchase order is entered into the contract database. Using information from the database, the contracting officer reviews the terms and conditions of the contract and determines whether the purchase order is valid. The purchase order is valid if the contract has not expired, the type of equipment ordered is listed on the original contract, and the total cost of the equipment does not exceed a predetermined limit. If the purchase order is not valid, the contracting officer sends the purchase order back to the requesting agency with a letter stating why the purchase order cannot be filled, and a copy of the letter is filed. If the purchase order is valid, the contracting officer enters the purchase order number into the contract database and flags the order as outstanding.

The purchase order is then sent to the Order Fulfillment department. Here the inventory is checked for each item ordered. If any items are not in stock, the Order Fulfillment department creates a report listing the items not in stock and attaches it to the purchase order.

All purchase orders are forwarded to the warehouse, where the items in stock are pulled from the shelves and shipped to the customer. The warehouse then attaches to the purchase order a copy of the shipping bill listing the items shipped and sends it to the contracting officer. If all items were shipped, the contracting officer closes the outstanding purchase order record in the database. The purchase order, shipping bill, and exception report (if attached) are then filed in the contracts office.

- (a) Develop a context diagram to represent the contracting system for GSC as described in the narrative.

[5 marks]

- (b) Draw a **LOGICAL** level-0 data flow diagram for GSC system.

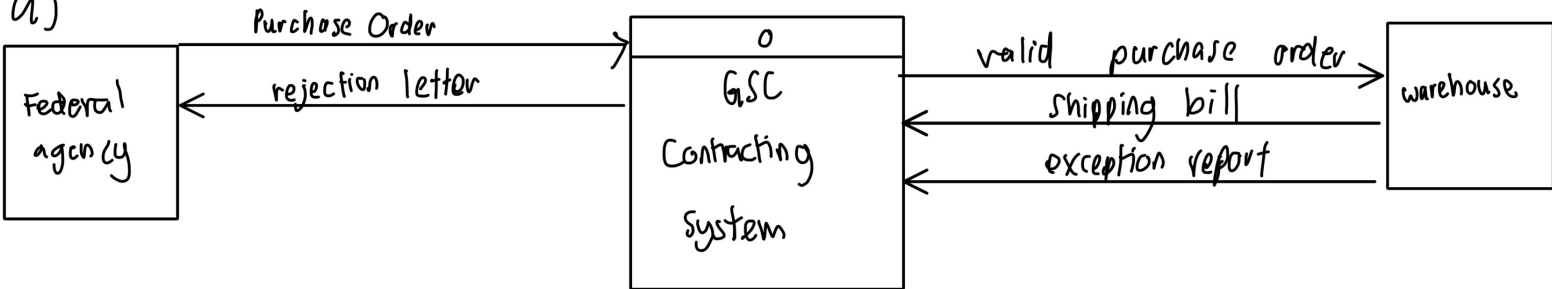
[15 marks]

- (c) Draw a **PHYSICAL** level-1 data flow diagram for ONE of the main process that you have identified in your answer in (b).

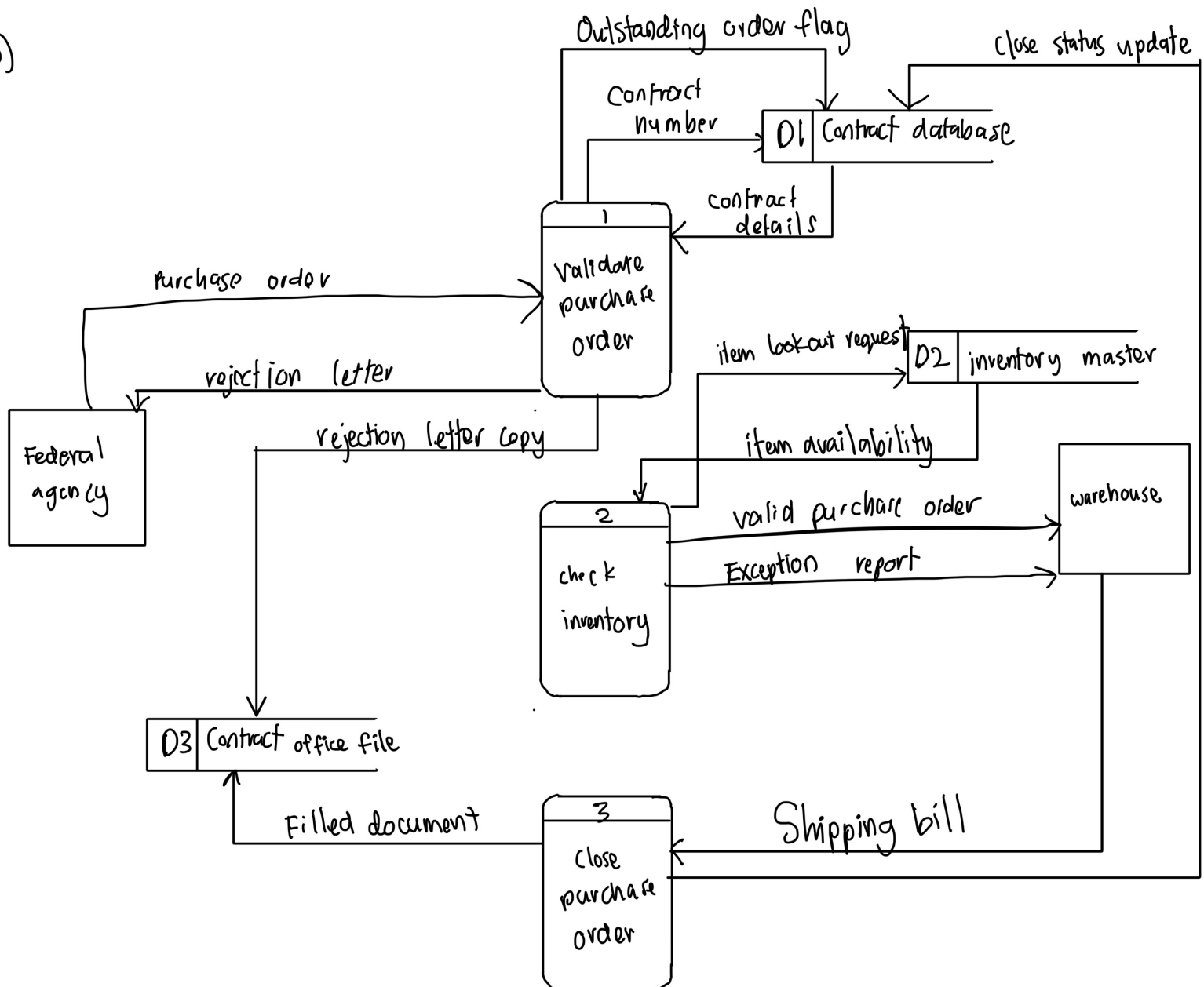
[10 marks]

Case Study

a)



b)



c) Physical child diagram process I

